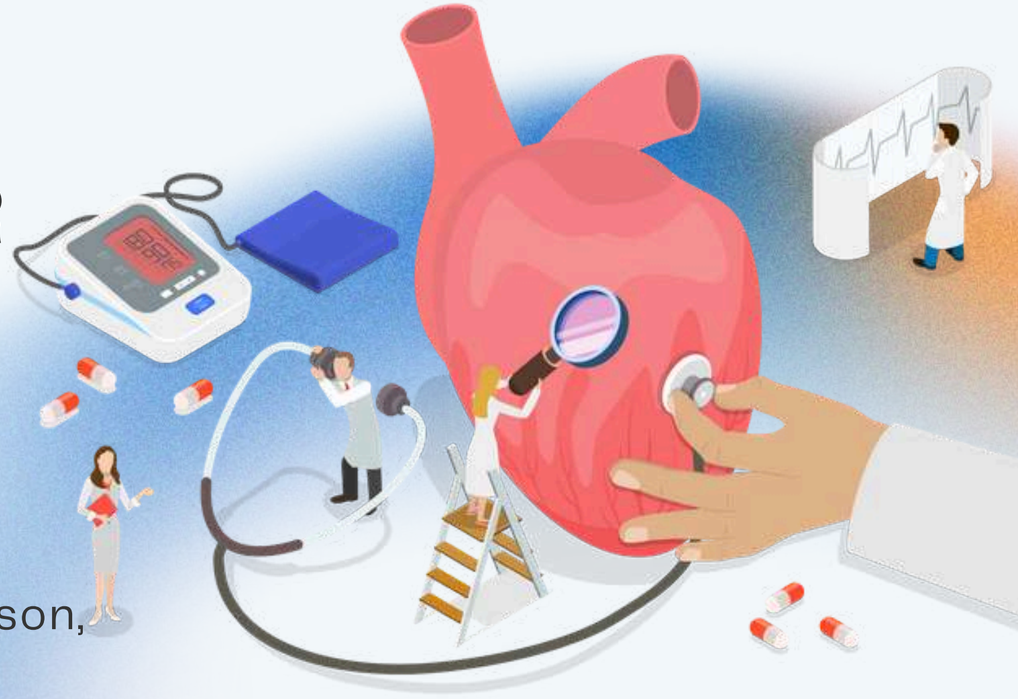
MOSAIX

INNOVATIVE PLATFORMS FOR HIGH-RESOLUTION CELLULAR ELECTROPHISIOLOGY



Presented By: Tomas Jonmundsson, Charlotte Leycuras, Jack Allinson, Trinity Tay, Tanisha Majumdar, Kaiyun Luo, Indra Malik

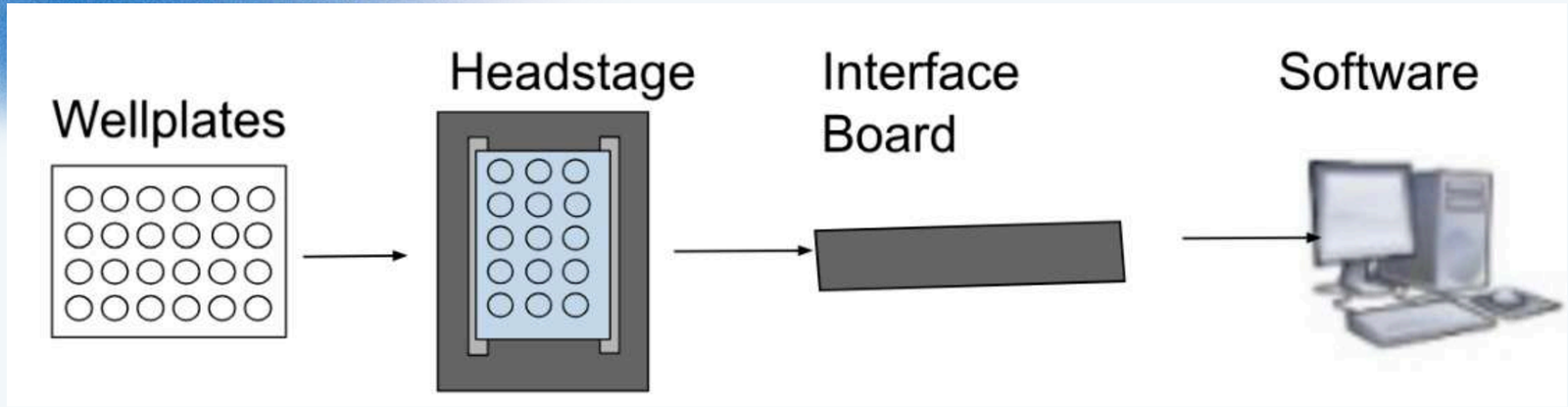
Clinicians: Dr Poonam Balalji and Dr Edward Yang

WESTMEAD

NSW GOVERNMENT Health Western Sydney Local Health District

[1]

CLINICAL PROBLEM



[2]

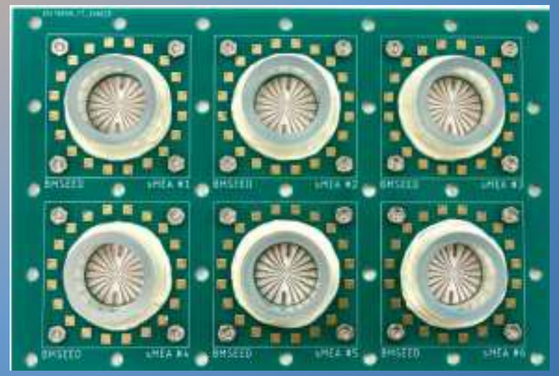
CURRENT MARKET

OTHER MEA SYSTEMS

1 Standard 24 Well Plates

[3]

2 6-Well Modular Plates



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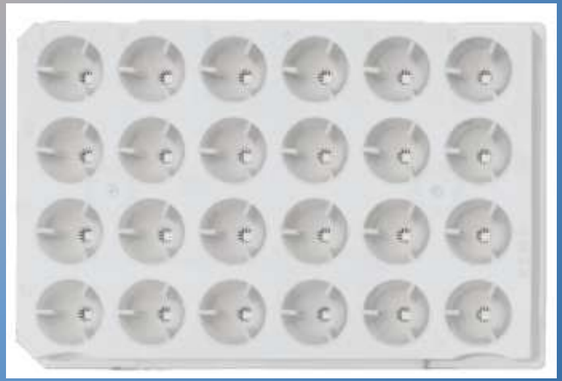
3 Hybrid/Optical Plates



[7]

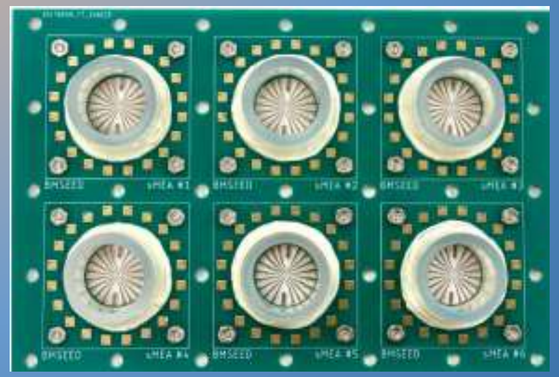
CURRENT MARKET

OTHER MEA SYSTEMS

1 Standard 24 Well Plates

- ✓ MODERATE SPATIAL RESOLUTION [3]
- ✗ NO CONTRACTILITY/OPTICAL MEASUREMENTS
- ✗ SMALL RECORDING AREA

2 6-Well Modular Plates



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3 Hybrid/Optical Plates

[7]

CURRENT MARKET

OTHER MEA SYSTEMS

1 Standard 24 Well Plates

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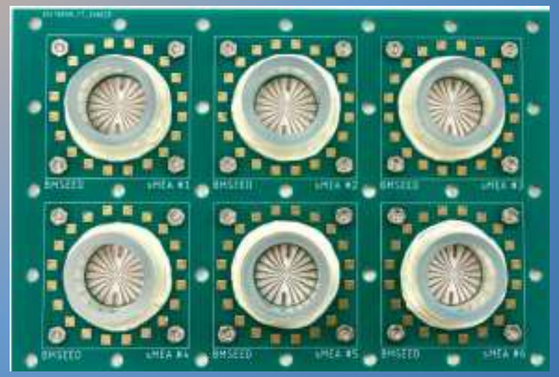
2 6-Well Modular Plates



[4-6]

- ✓ MIMICS PHYSIOLOGY
- ✗ POOR SPATIAL RESOLUTION
- ✗ CONSUMABLE PLATES

3 Hybrid/Optical Plates

[7]

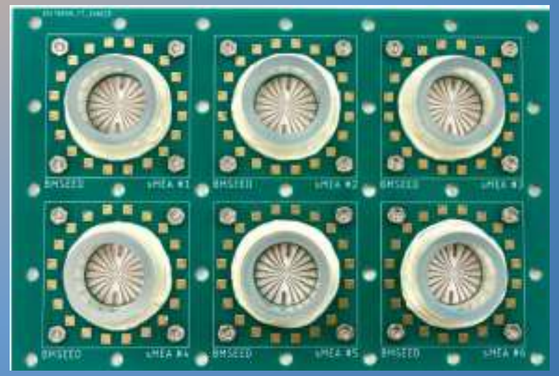
CURRENT MARKET

OTHER MEA SYSTEMS

1 Standard 24 Well Plates

[3]

2 6-Well Modular Plates



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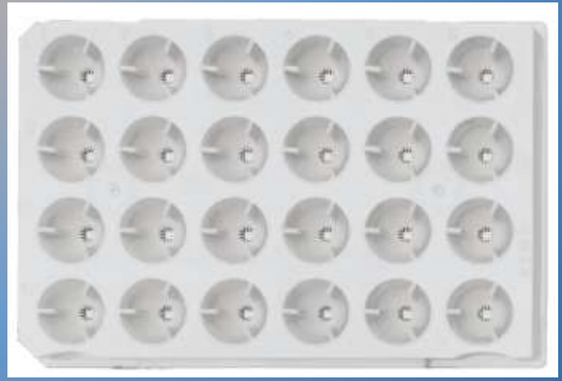
3 Hybrid/Optical Plates



[7]

- ✓ SPATIAL RESOLUTION
- ✓ ORGANOID COMPATIBLE
- ✗ SMALL RECORDING AREA

OUR SOLUTION

[1]

HOW DOES IT WORK ?

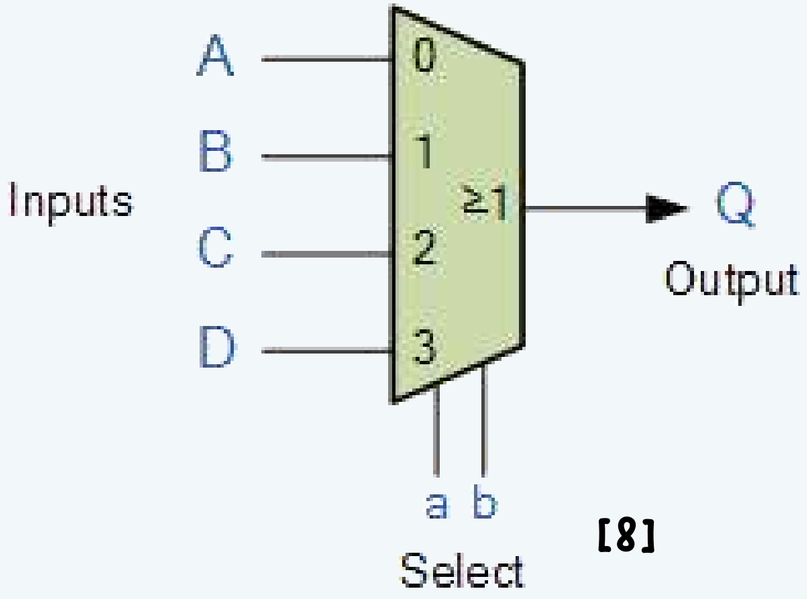
WELL 1

WELL 2

WELL 3

WELL 4

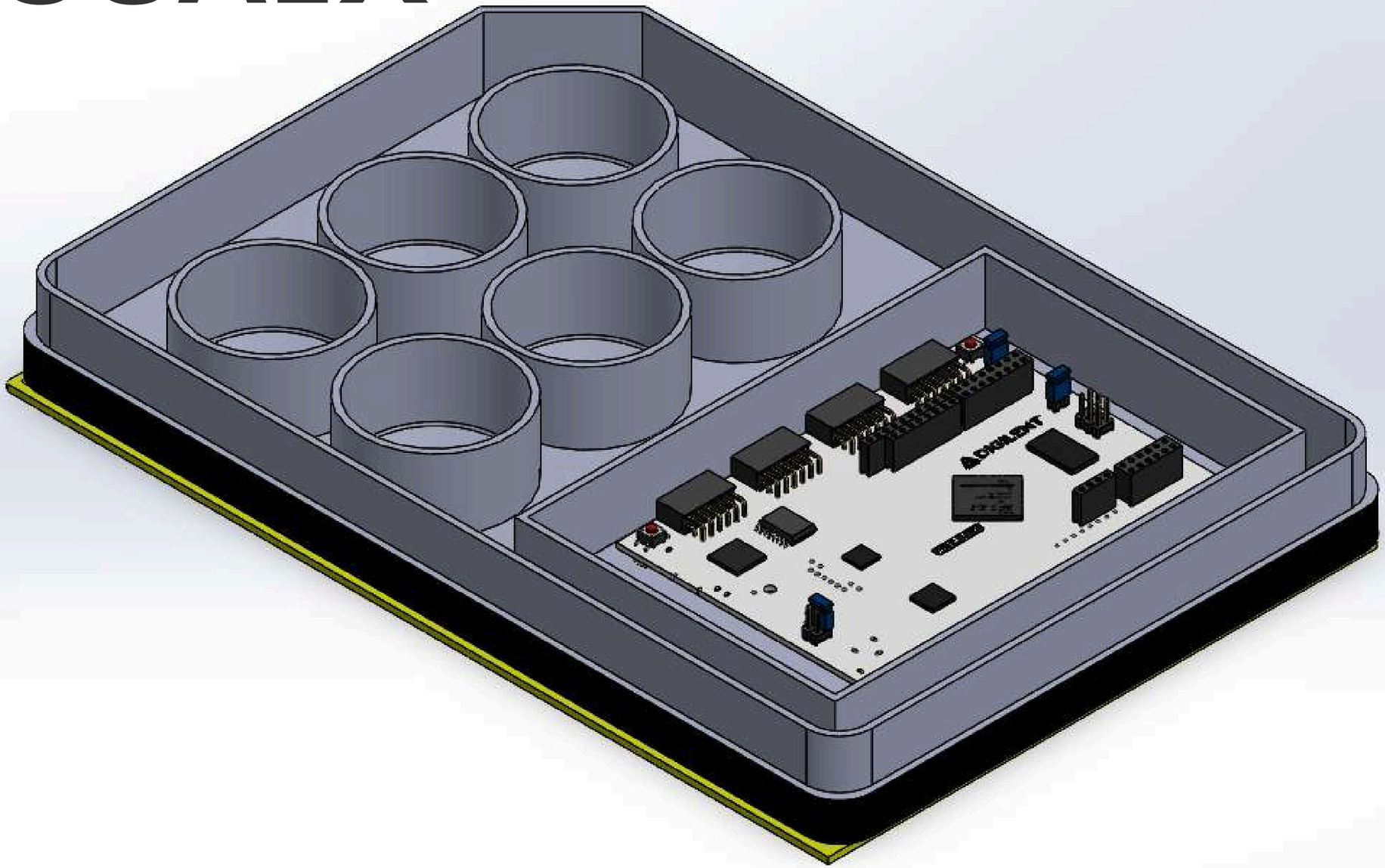
MULTIPLEXOR



[8]

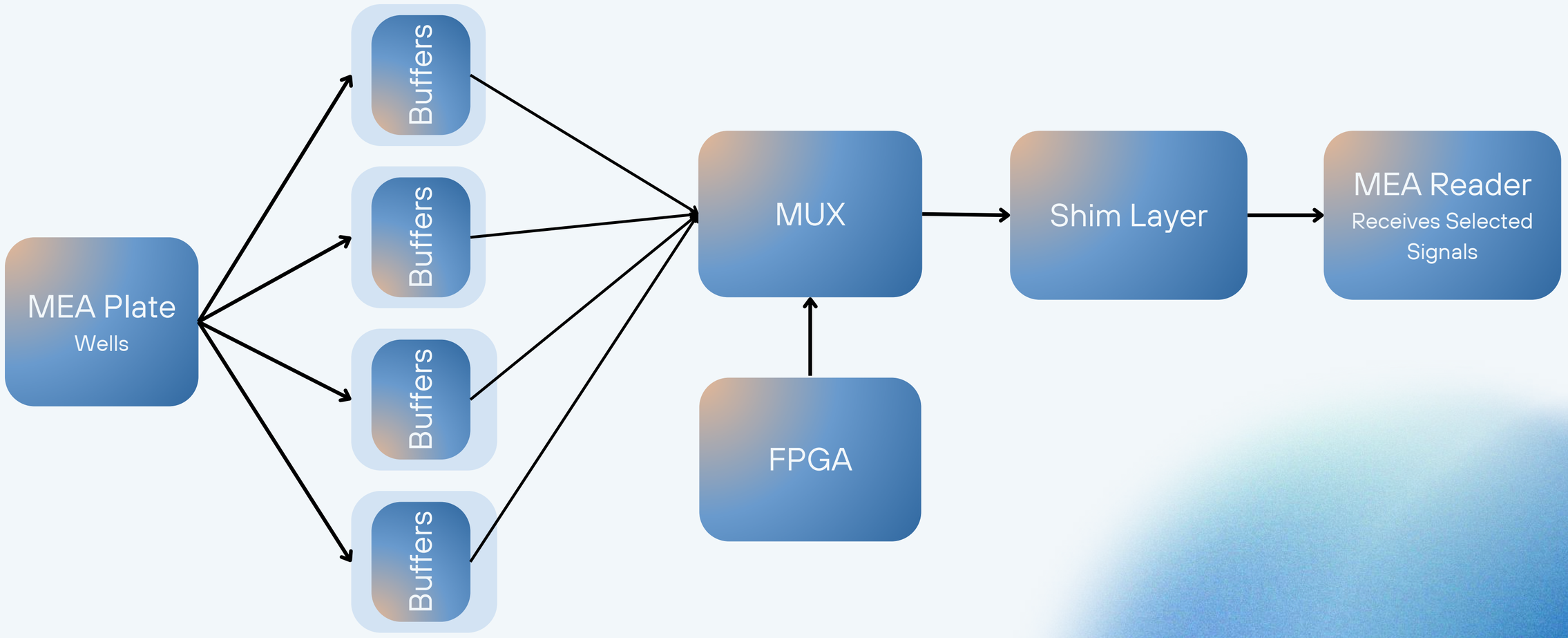
FUNNELING THROUGH SELECTED WELL

MOSAIX



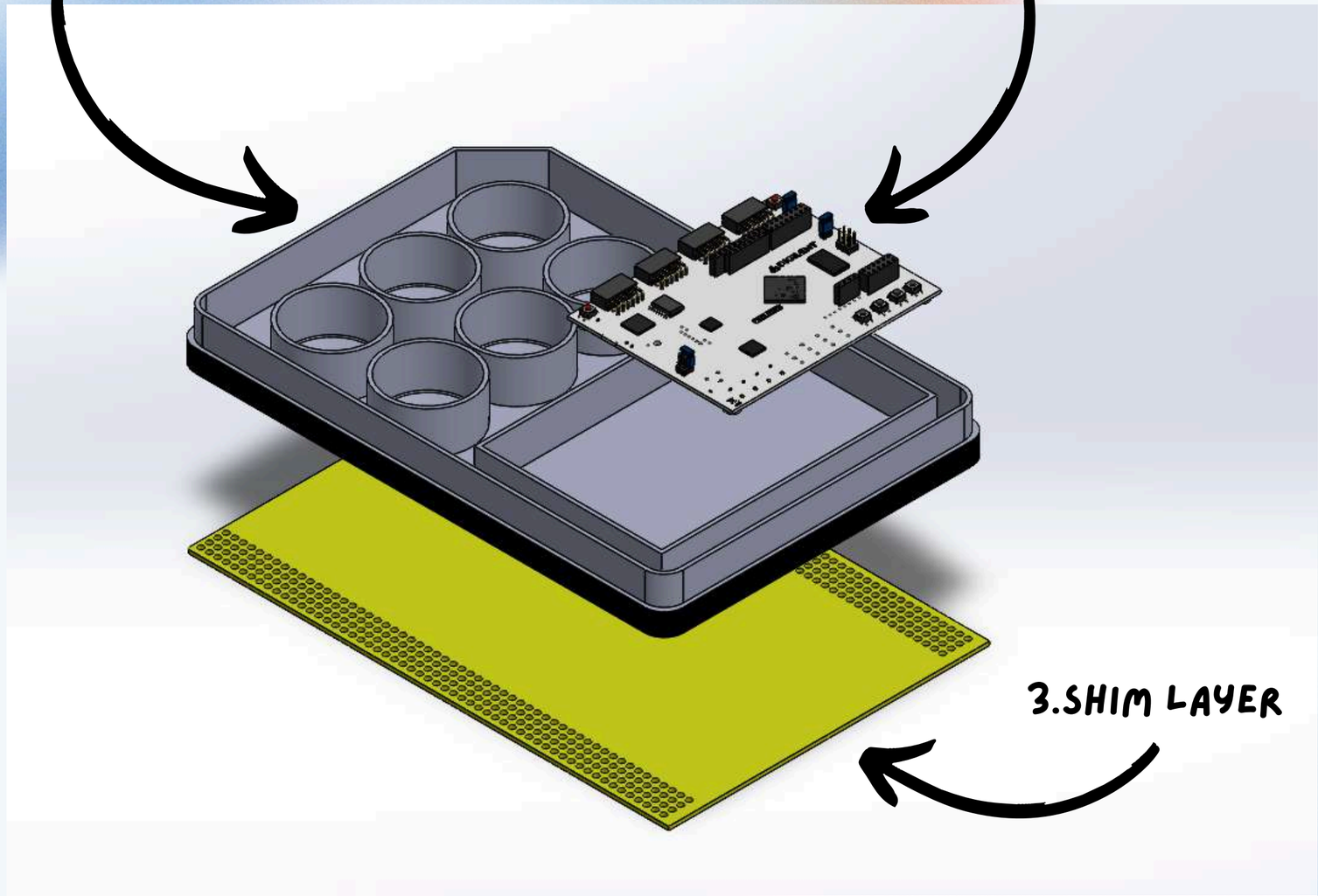
DESIGN SOLUTION

BLOCK DIAGRAM



1. MEA PLATE

2. MULTIPLEXOR-FPGA PCB



3. SHIM LAYER

4. MEA READER

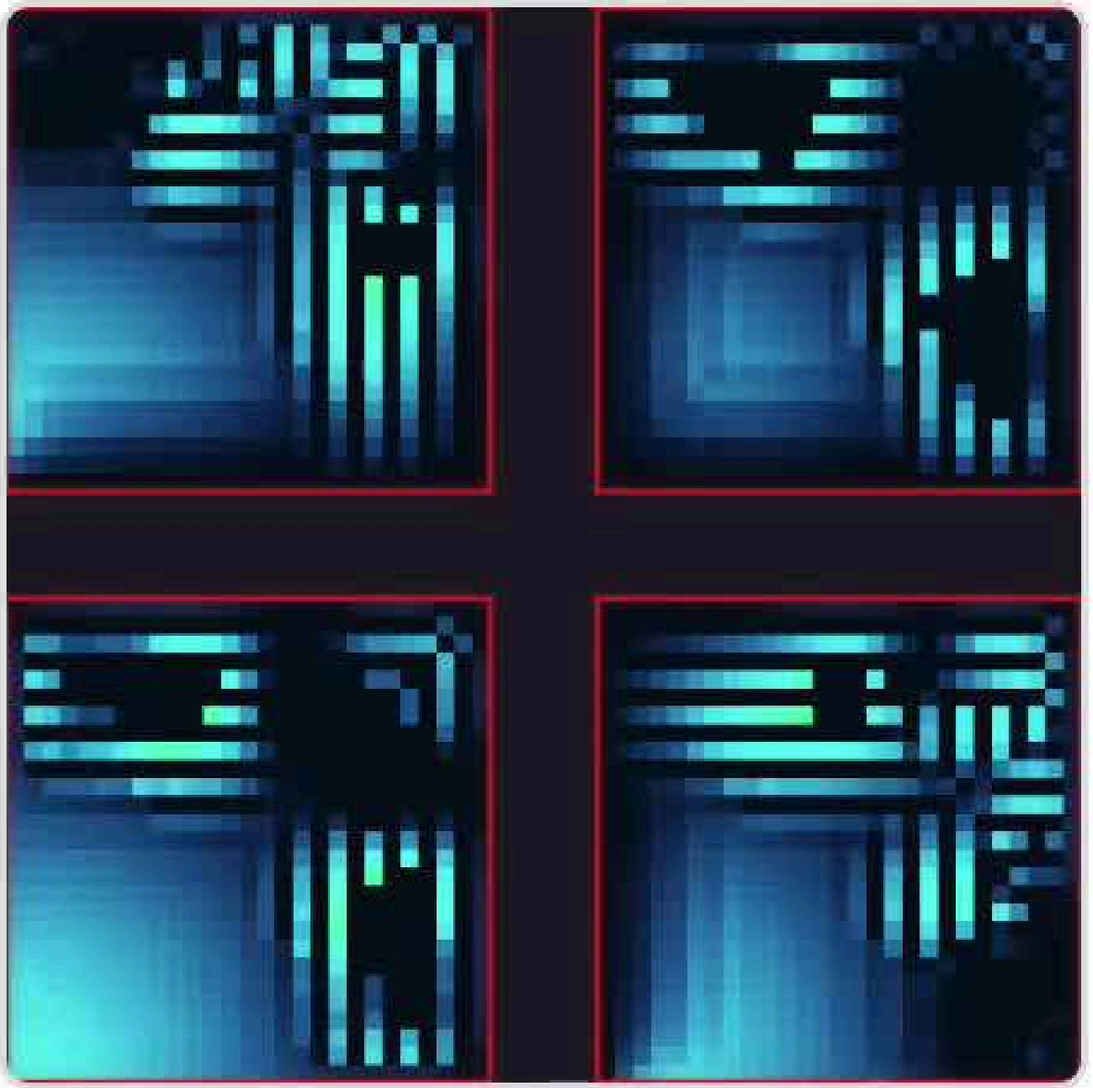
[9]

5. PROPRIETARY USER INTERFACE

[10]

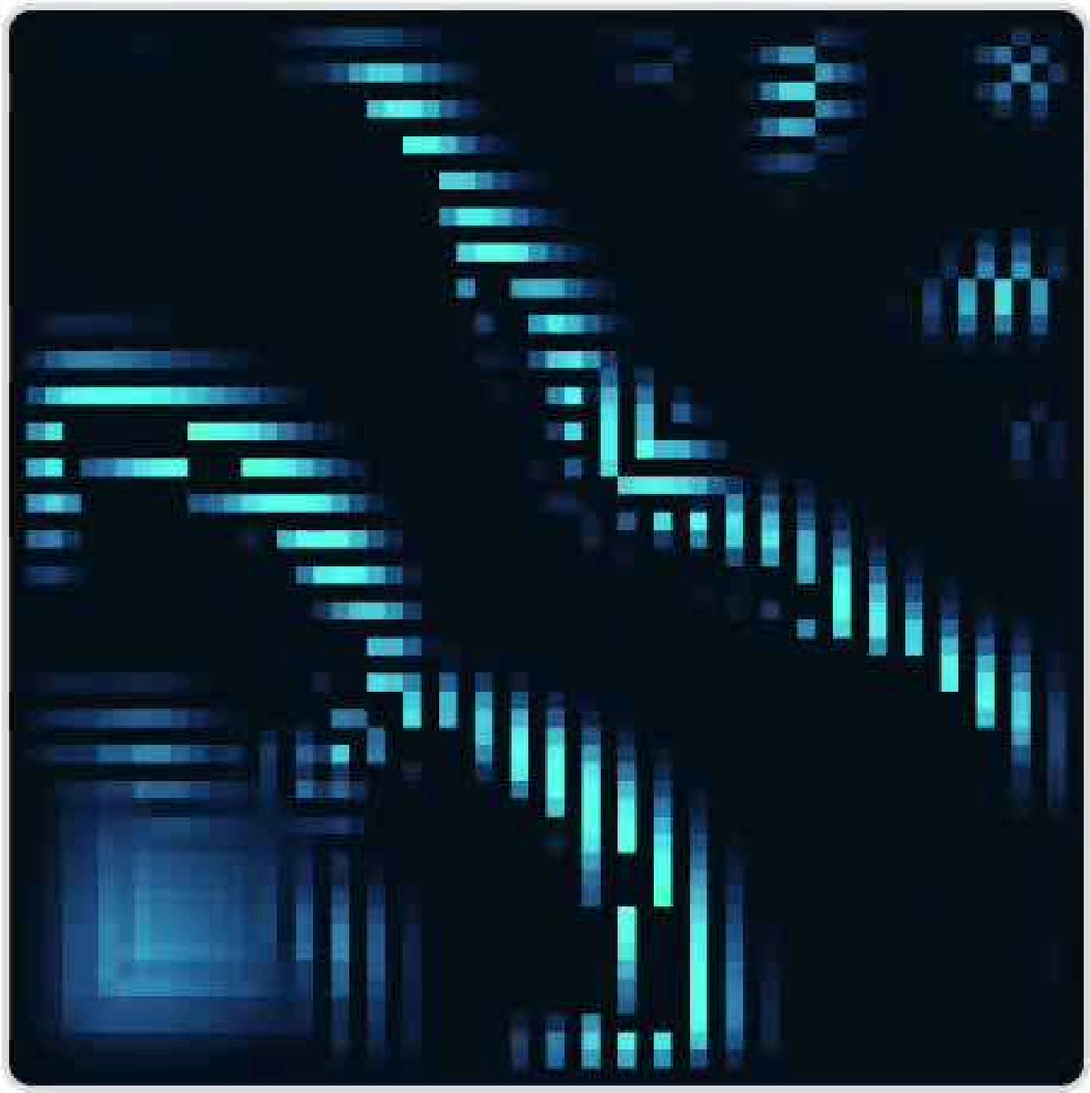
Current MEA Plate

View of 4 Wells



MEA Plate With Mosaix

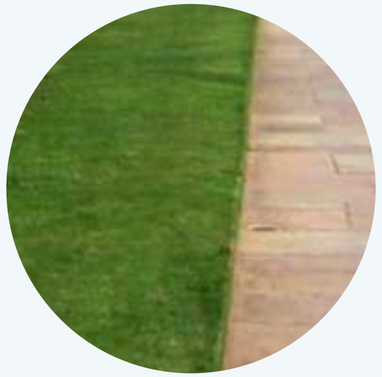
View of 4:1 Multiplexed Well



INNOVATION

WHY ARE WE BETTER THAN THE REST?

SPATIAL RESOLUTION



INNOVATION

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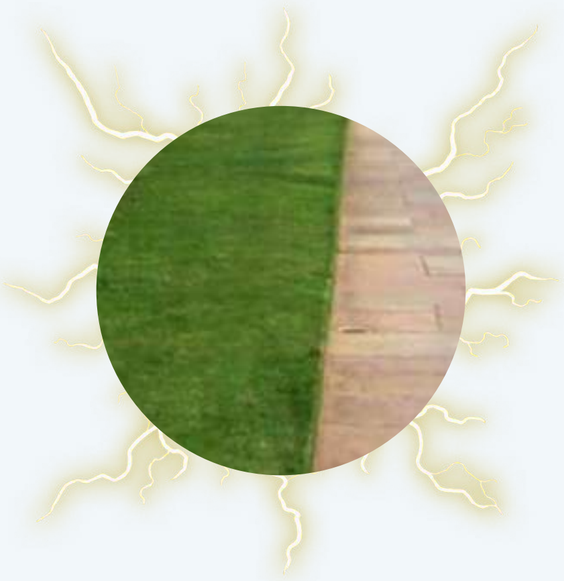
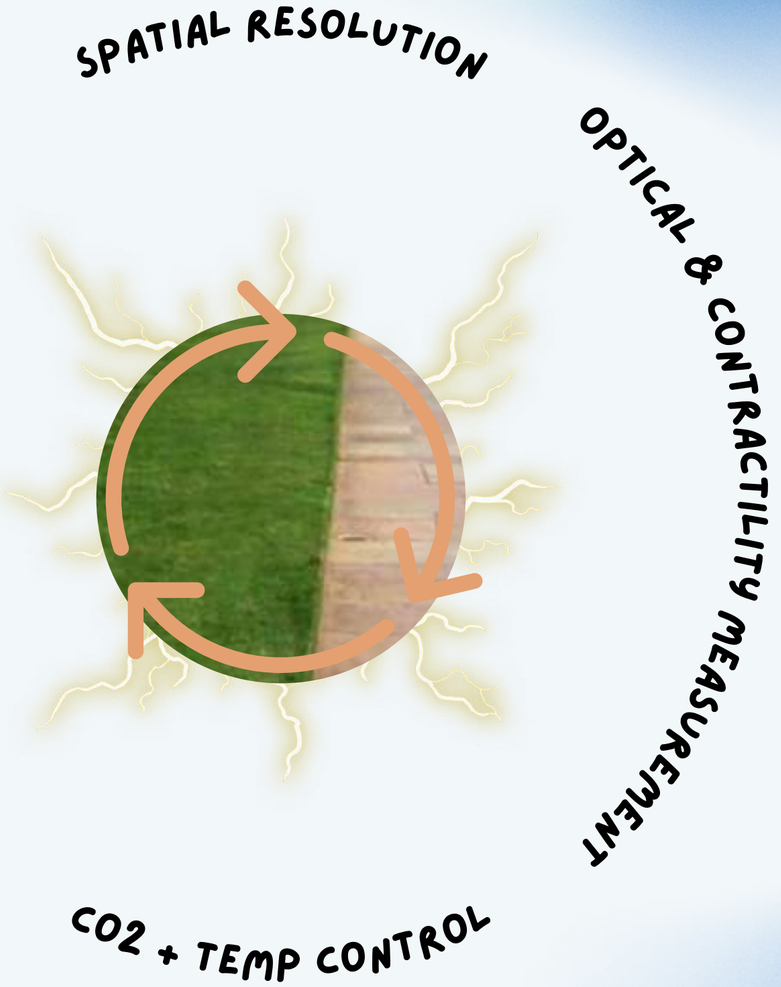
SPATIAL RESOLUTION



OPTICAL & CONTRACTILITY MEASUREMENT

INNOVATION

WHY ARE WE BETTER THAN THE REST?



INNOVATION

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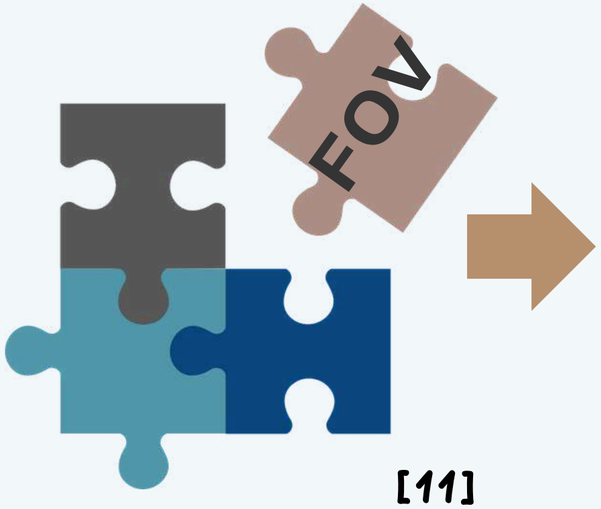

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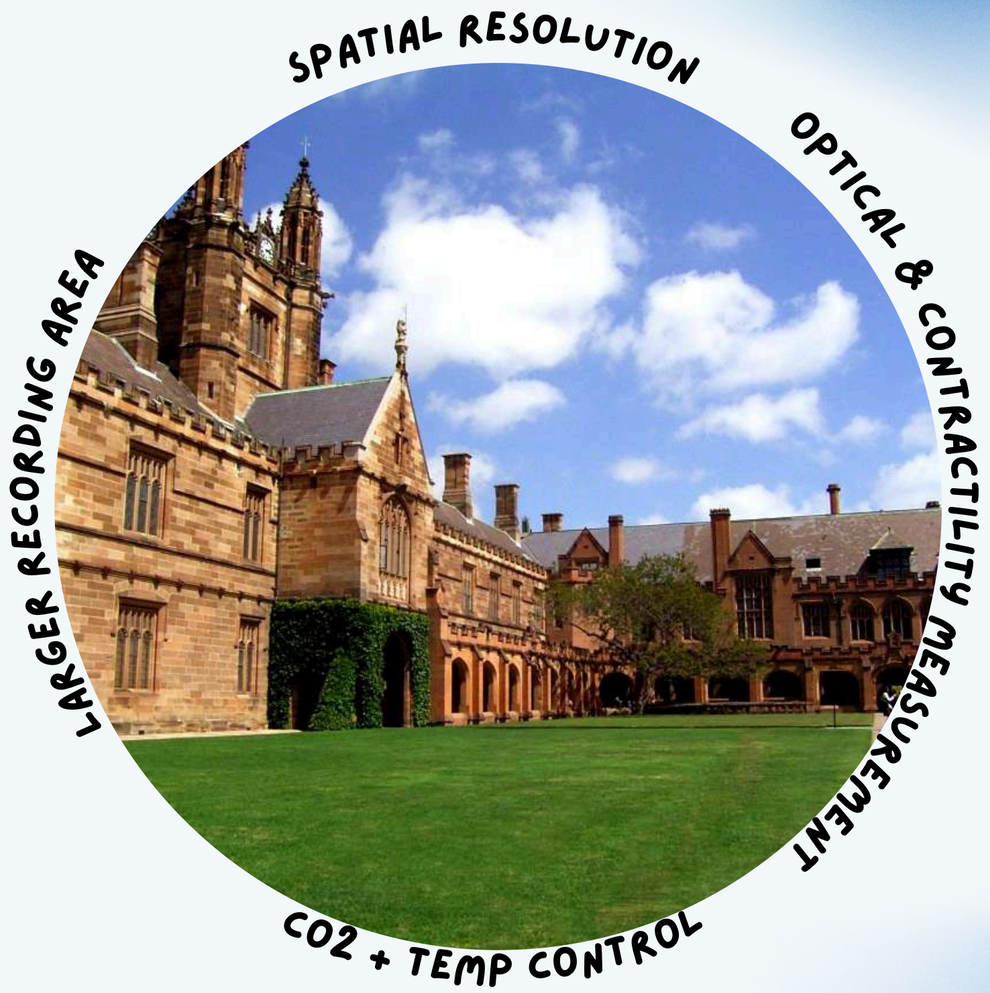
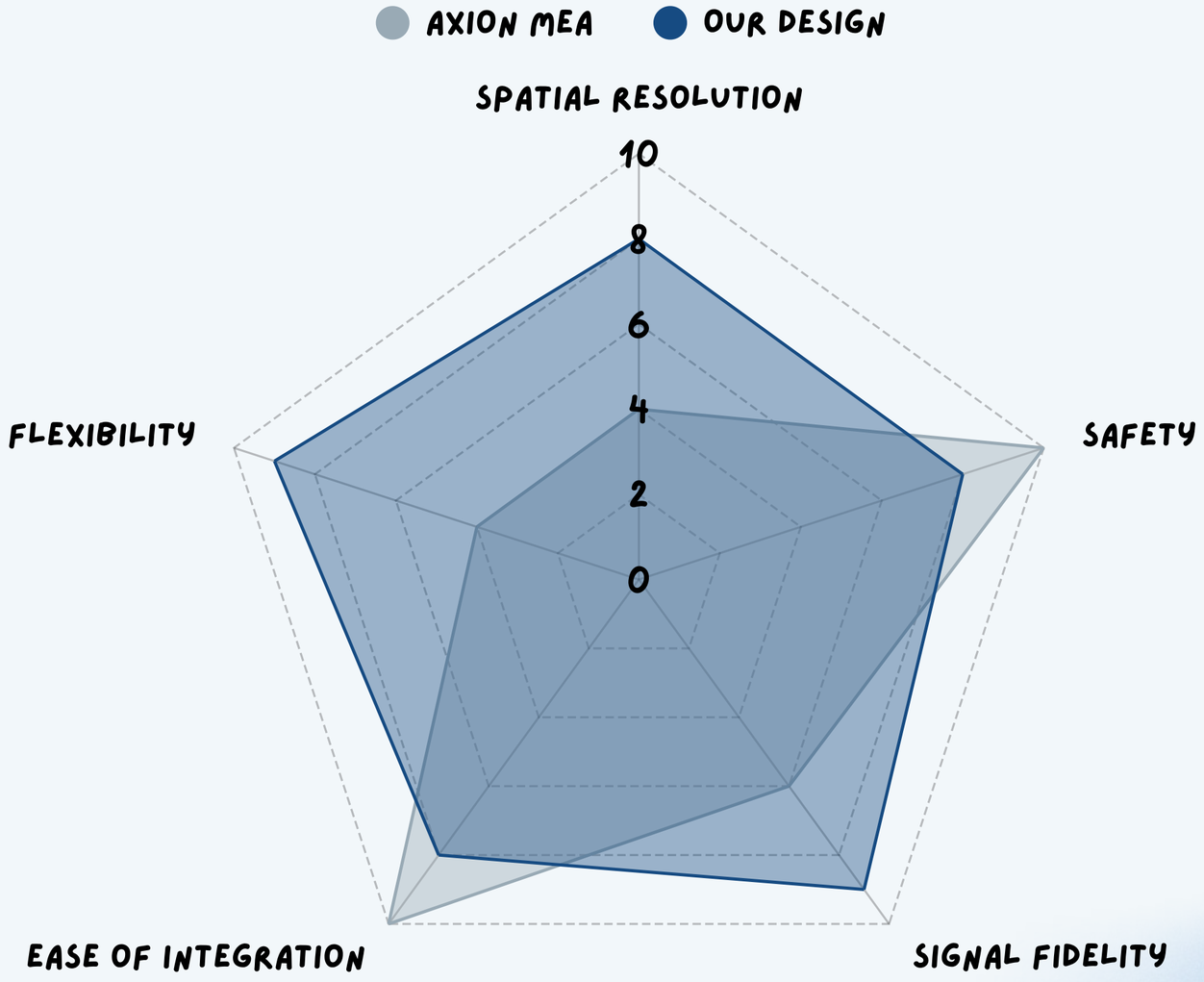


FEASIBILITY



WHAT NEXT?

Scaling up!

QUESTIONS?

THANKS FOR LISTENING

REFERENCES

Acknowledgement of AI Use
Generative AI (i.e., ChatGPT-5, Microsoft Copilot, Claude Sonnet 4.5) was used to generate some images in order to enhance clarity of the explanation.

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